Survey Paper Assistant: A Retrieval-Augmented Generation (RAG) Framework for First-Time Academic Writers

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Abstract

Writing an academic survey or comprehensive literature review presents significant cognitive and pedagogical challenges for undergraduate and early-stage postgraduate students. Novice researchers frequently struggle to formulate structured domain taxonomies, synthesize divergent findings across multiple technical documents into a cohesive narrative, avoid unintended paraphrasing/copying, and objectively assess draft completeness prior to peer submission. To address these bottlenecks, this project proposes the **Survey Paper Assistant**, an intelligent, retrieval-augmented conversational web platform designed to guide first-time academic writers systematically from raw source documents to an evaluation-ready survey draft.

The platform is architected around a modular Retrieval-Augmented Generation (RAG) pipeline tailored specifically for academic corpus workflows. Key system capabilities include: **(i) Multi-Document Ingestion & Extraction**, which processes batch academic PDFs, parses metadata, extracts section structures, and gracefully isolates corrupted or non-extractable files; **(ii) Grounded Taxonomy & Outline Generation**, constructing customizable survey skeletons directly rooted in thematic clusters of the ingested corpus rather than generic templates; **(iii) Hierarchical Summarization**, providing both granular per-paper extractions (problem, methodology, findings, and limitations) and cross-document syntheses that capture consensus, controversy, and open literature gaps with provenance backlinks; **(iv) Automated Draft Review & Diagnostics**, executing multi-point inspections on structural completeness, logical transitions, citation density, readability, and localized verbatim overlap against the reference corpus; and **(v) Sandboxed Conversational Guidance**, facilitating iterative contextual queries over project artifacts through memory-efficient context-window management. By enforcing interactive scaffolding and rigorous verification gates, the system democratizes academic writing literacy, enhances scholarly rigor, and shortens the trajectory from raw research literature to high-quality survey publications.

Keywords: Retrieval-Augmented Generation (RAG), Academic Writing Assistant, Survey Paper Synthesis, Multi-Document Summarization, Automated Draft Review, Natural Language Processing.

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1 Problem Statement & Motivation

Academic survey papers serve as critical foundations for scientific progress, providing structured overviews, taxonomies, and critical analyses of evolving research fields. However, students assigned to compose their first survey paper (such as final-year B.Tech and graduate students) face acute cognitive hurdles:

- **Taxonomic Ambiguity:** Difficulty in discerning conceptual categories and categorizing diverse research methodologies into coherent taxonomies.
- **Synthesis Deficits:** Inability to cross-examine related papers, resulting in disconnected per-paper summaries rather than an integrated, critical narrative.
- **Academic Writing Conventions:** Lack of intuition regarding section hierarchies, transitions, and proper citation anchoring for technical claims.
- **Pre-Submission Anxiety:** Absence of objective, iterative feedback mechanisms to audit draft completeness and inadvertent verbatim overlap before formal evaluation.

2 System Architecture & Proposed Pipeline

The system bridges the gap between an unorganized repository of source PDFs and a structured manuscript through five decoupled, coordinated subsystems:

1. **Project-Scoped Data Management:** Enforces complete isolation between separate research topics. Each workspace persists uploaded files, generated outlines, hierarchical summaries, diagnostic logs, and conversational state.
2. **Document Ingestion & Anomaly Filtering:** Utilizes parallel PDF text and layout extraction, identifying metadata (title, authors, abstract, section headings) while detecting edge-case anomalies (scanned image PDFs, password restrictions, non-English text, and duplicate uploads).
3. **Grounded Structure Generator:** Analyzes semantic embeddings and topic models derived from the ingested papers to synthesize an outline grounded in literature evidence (e.g., background, classification taxonomy, comparative tables, and open challenges) with interactive reordering and steering capabilities.
4. **Hierarchical Synthesized Summarization:** Produces standardized single-paper analytical cards alongside an aggregated multi-document matrix that highlights consensus, conflicting findings, and unexplored research frontiers, maintaining bi-directional citation traceability.
5. **Diagnostic Draft Review Engine:** Audits uploaded drafts (PDF/DOCX/text) across structural completeness, argumentative flow, citation presence, readability indices, and cosine/n-gram similarity against the primary sources to prevent unintentional plagiarism.
6. **Project-Scoped RAG Chatbot:** An interactive conversational agent equipped with project-level memory, contextual compression, and document grounding to address specific synthesis inquiries without hallucinations.

3 Functional Overview

4 Evaluation & Expected Impact

The effectiveness of the Survey Paper Assistant is benchmarked across functional reliability, content fidelity, and pedagogical usability:

- **Performance Targets:** End-to-end ingestion and summarization for standard papers (≤ 10 pages) within 60 seconds, with scalable concurrent support for ≥ 6 documents per project.

Module	Core Functionality	Target Deliverable / User Benefit
FR-1: Management	Multi-project state persistence	Seamless workspace switching without session loss.
FR-2: Ingestion	Asynchronous batch PDF processing	Extracted metadata, fault handling (scans/corruptions).
FR-3: Structuring	Grounded outline generation	Evidence-based taxonomy, drag-and-drop hierarchy.
FR-4: Synthesis	Single & multi-paper summarization	Cross-paper synthesis matrix with source provenance.
FR-5: Review	Multi-criteria draft auditing	Pinpointed issue diagnostics, severity, and suggested fixes.
FR-6: Dialogue	Isolated RAG conversational thread	Contextual Q&A with adaptive window compression.
FR-7: Export	Multi-format generation	Exportable outlines and reviews (PDF, DOCX, Markdown).

Table 1: Summary of primary functional modules defined in the system PRD.

- **Content Quality:** Evaluated via human rubrics assessing factual accuracy, narrative coherence, absence of hallucination, and taxonomy relevance.
- **Usability Metrics:** Measured by time-to-first-outline, reduction in iterative draft review cycles, and user confidence scores among first-time student authors.